

REVIEW

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Machine learning applications in energy systems: current trends, challenges, and research directions

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Abstract

The paradigm shift towards Smart Grids, Smart Buildings, Smart Monitoring, and Operation has driven researchers to propose innovative solutions for designing and maintaining energy systems. Although the integration of Renewable Energy Sources (RES) supports sustainability goals, it also introduces vulnerabilities to unpredictable challenges such as grid stability, energy storage requirements, and infrastructure modernization. Machine Learning (ML) has emerged as a transformative tool to address these challenges, offering opportunities to enhance energy efficiency, and system design in alignment with Sustainable Development Goals (SDGs). The emphasis on these goals necessitates the study of new system designs that prioritize energy efficiency. Building on its proven success, researchers are increasingly adopting ML-driven approaches to accelerate advances in energy systems. This work presents a detailed review of current ML-driven research trends in energy systems, outlines the associated challenges, and provides potential research directions and recommendations. Unlike the existing literature, which focuses primarily on ML applications in the RES domain, this study offers a holistic perspective on ML-driven approaches across various aspects of energy systems, including energy policy and sustainability. It aims to serve as a comprehensive resource, bridging the gap between research advancements and practical implementations in energy systems through ML-driven innovation.

Keywords Machine learning, Renewable energy systems, Sustainable energy, Energy efficiency, Smart grids

Introduction

Emerging economies are experiencing rapid growth in energy consumption while facing increasing pressure to enhance energy efficiency and reduce carbon emissions. Despite extensive research on energy-saving mechanisms, continuous advancements in the energy sector pose persistent challenges. For instance, the integration of Renewable Energy Sources (RES) disrupts traditional energy and power management due to their intermittent nature, mainly influenced by the variability of solar and wind energy

[1]. Conventional power grids, originally designed for stable energy sources, struggle to accommodate these fluctuations, necessitating innovative solutions.

The digital transformation of energy systems has introduced new technologies such as real-time monitoring, advanced metering, smart grids, and blockchain-based energy transactions. These innovations can significantly improve system resilience and efficiency, yet their integration must align with global sustainability objectives. The achievement of energy-efficient systems is essential to ensure a more reliable and adaptable energy future.

Artificial Intelligence (AI), particularly Machine Learning (ML), has emerged as a powerful tool to address the complexities of modern energy systems. The energy infrastructure generates vast amounts of data, from grid operations to consumption patterns, providing a valuable foundation for ML applications. Unlike traditional models, ML excels in analyzing large, dynamic datasets to uncover patterns, predict energy demands, and optimize resource allocation [1]. By improving renewable energy forecasting, load distribution, and device automation, ML plays a crucial role in minimizing energy waste and enabling data-driven decision making for sustainable energy management.

Recent advances in ML, coupled with the proliferation of IoT devices and improved data collection techniques, have further amplified its potential [2]. Real-time data acquisition allows ML models to learn and adapt to changing energy environments, enhancing predictive capabilities for system performance, energy storage management, and demand-side optimization. Although ML has been successfully deployed in various engineering applications [3], its role in energy systems remains underdeveloped. Unlike in fields such as text mining or natural language processing, where ML has led to independent disciplines, its application in engineering primarily serves as an optimization and analytical tool [2, 3]. By simplifying complex mathematical models and reducing computational burdens, ML can facilitate more efficient energy management and operations.

Despite its potential, the integration of ML in the energy sector has lagged behind its widespread adoption in other fields. Despite the fact that ML has been successfully applied to various energy-related challenges, such as forecasting wind power and optimizing energy distribution [4], its full potential remains underexplored. Recent advancements suggest that ML can significantly improve the efficiency, reliability, and sustainability of energy systems. In particular, ML is expected to play an essential role in green electricity generation, demand side management, and predictive maintenance, thus reducing the carbon footprint compared to conventional techniques [5]. This article provides a comprehensive review of ML applications in the energy sector, emphasizing their role in optimizing energy generation, distribution, and storage while addressing challenges related to the integration of renewable energy. Additionally, we discuss the implications of ML for energy policy and economic decision making. By identifying key research gaps and offering actionable recommendations, this study aims to guide future research toward the development of AI-driven, energy-efficient systems aligned with sustainability goals.

The contributions of this work are as follows:

1. This research provides a comprehensive overview of recent advancements in ML applications for energy systems. Unlike most existing reviews, which focus primarily on Renewable Energy Sources (RES), this work presents a broader perspective by covering diverse ML-driven applications across various aspects of energy systems. By

synthesizing insights from multiple domains, this study highlights emerging trends and the evolving role of ML in optimizing energy management, grid operations, and predictive analytics.

2. A detailed discussion is provided on the effectiveness and challenges of implementing ML in the energy sector. This work examines data availability, model interpretability, real-time decision making, and scalability issues, which remain significant barriers to widespread ML adoption. In addition, it explores the trade-offs between accuracy, computational efficiency, and security in ML-driven energy solutions. By analyzing these challenges alongside emerging opportunities, this research provides valuable insights for overcoming key obstacles in the field.
3. This work identifies critical future research directions and offers practical recommendations to enhance ML applications in energy systems. It emphasizes the need for more interpretable, scalable, and energy-efficient ML models, as well as novel approaches such as transfer learning, federated learning, and physics-informed ML. Furthermore, this research highlights the potential of ML to drive sustainability, improve energy policy frameworks, and support the transition towards intelligent, low-carbon energy systems.

The organization of the manuscript is as follows: Section "[Introduction](#)" introduces the background study of ML applications in energy systems. Section "[Machine learning techniques in energy systems](#)" presents a comprehensive review of original research and review articles that explore the application of various machine learning (ML) techniques in energy systems. Section "[Research design](#)" outlines the research design, including the methodology used to select and categorize the reviewed literature. Section "[Applications of machine learning in the energy sector](#)" explores the diverse applications of ML across the energy sector, highlighting its impact on energy distribution, efficiency, storage, and smart grid technologies. Section "[Challenges of implementing ML in energy systems](#)" addresses the major challenges involved in implementing ML within energy systems. Section "[Recommendations and future research directions](#)" presents future research directions and provides recommendations for advancing ML-driven energy solutions. Finally, Section "[Conclusion](#)" concludes the manuscript by summarizing the key findings and emphasizing critical considerations for building sustainable and AI-powered energy systems. The overall structure of the manuscript is illustrated in Fig. 1.

Machine learning techniques in energy systems

This section presents a structured and in-depth review of recent state-of-the-art research on the application of machine learning (ML) techniques in energy systems. The reviewed studies were identified and selected through a rigorous systematic review process, as detailed in Section "[Research design](#)". By applying predefined search criteria and inclusion parameters, the review focuses on peer-reviewed journal articles and conference papers published between 2019 and 2025 that explicitly address ML driven solutions in energy-related domains.

In addition to core engineering applications, this section incorporates multidisciplinary perspectives by including works related to energy policy, economics, and system management, provided they demonstrate empirical integration of ML methodologies. The discussion spans a range of ML techniques from traditional algorithms to deep and hybrid models and highlights their implementation in key energy domains such as

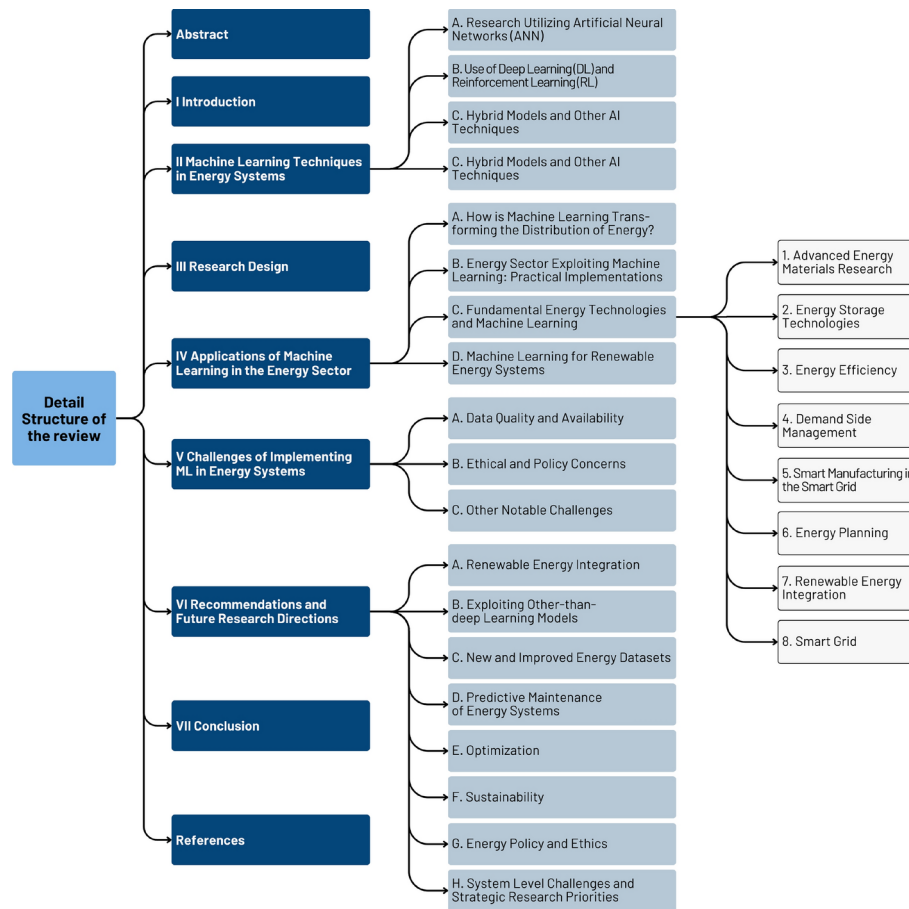


Fig. 1 Detail structure of the manuscript

prediction, optimization, forecasting, efficiency, and smart grid operations. The analysis emphasizes validated research outcomes, offering insights into both technical performance and strategic impact within the energy sector.

Research utilizing artificial neural networks

Artificial Neural Networks (ANNs) are foundational machine learning models composed of interconnected neurons organized in layers, enabling them to process and learn from complex data patterns [6]. Through forward propagation and backpropagation, ANNs iteratively adjust their weights to minimize errors and improve predictive accuracy. While their relatively simple architecture makes them effective for various applications, including energy systems, their limitations in handling complex hierarchical features have led to the development of deeper architectures. These deep ANNs enhance the modeling of nonlinear relationships in energy-related problems, such as demand forecasting, optimization of renewable energy integration, and intelligent power distribution. In recent years, there have been several notable research studies that have applied ANNs to optimize processes and improve efficiency in energy-related systems.

Rangel-Martinez et al. [2] provided a comprehensive discussion on ML applications in the manufacturing sector that have a significant impact on sustainability and the environment, including RES. This study recommends ANNs to optimize manufacturing processes. The authors anticipated that data collection, administration, and security will

play crucial roles in ML-based approaches. The algorithms developed are expected to be accessible to stakeholders in the energy sector, thereby encouraging ambitious energy projects. New algorithms that can produce dependable data and add new information sources are expected to improve the information flow between ML and associated systems. Unsupervised and reinforcement learning (RL) will be significant for the energy sector, but this will depend on advancements in related fields such as data science and big data analytics. Although the study is primarily qualitative, it refers to performance improvements in production processes by as much as 12 to 18 percent through ANN-based optimization in pilot testbeds.

The research work by B. Adibimanes et al. explores the application of machine learning algorithms to improve the efficiency of energy production and consumption in sewage sludge incineration processes [7]. The authors used ANN and other techniques to develop a predictive tool which can optimize important operational parameters, saving energy significantly. Their model, validated using a dataset of more than 10,000 process entries, demonstrated an energy efficiency improvement of approximately 15 percent and reduced sludge disposal costs by up to 22 percent. Their research resulted in improving the energy performance of wastewater treatment plants and lowering the cost associated with sludge disposal and contributed to sustainable waste management practices. This demonstrates ANN's potential not just in energy forecasting, but in tangible energy savings and operational optimization in complex industrial processes.

ML can also be used in the energy storage sector to predict the thermal properties of green solvents. T. Lemaoui et al.'s study presents a groundbreaking integration of molecular modeling and machine learning to address the challenge of mapping thermal conductivity in ionic liquids (ILs) and deep eutectic solvents (DESs) [8]. By utilizing ANN, the authors developed a tool that accurately predicts the thermal properties of these green solvents, which are crucial for energy storage applications. The model achieved a mean absolute error of less than 0.04 W/mK across 120 solvent samples, indicating high predictive accuracy. This approach enhances the understanding of ILs and DESs by paving the way for their practical industrial use, offering a significant contribution to the field of sustainable energy. Their methodology combines computational chemistry with ML, exemplifying how hybrid modeling approaches can accelerate material discovery in next-generation energy storage systems.

In their 2023 study, Harrison et al. [9] introduced a robust nonlinear Maximum Power Point Tracking (MPPT) controller for photovoltaic (PV) systems, integrating Particle Swarm Optimization (PSO)-based integral backstepping with Artificial Neural Network (ANN) techniques. The proposed controller establishes a closed-loop system that includes the PV module, boost converter, load, an ANN model for reference prediction, and an integral backstepping controller. Stability is ensured through Lyapunov theory, while PSO optimizes the controller parameters. Numerical simulations demonstrated the controller's superior performance over traditional methods like Perturb and Observe (P&O) and PSO-terminal sliding mode controllers, particularly in reducing oscillations around the maximum power point and enhancing robustness. Experimental validations under real environmental conditions further confirmed the controller's efficacy in improving PV system efficiency and reliability.

The authors in [10] propose a physics-informed graphical learning state estimation method to enhance power system state estimation by integrating domain physical

knowledge with graphical neural networks. This approach aims to address limitations in traditional state estimation methods by leveraging both the physical laws governing power systems and the structural information captured by graphical models. The integration of physics-informed principles ensures that the neural network's predictions adhere to known system behaviors, thereby improving accuracy and reliability. The study demonstrates that this method effectively enhances state estimation performance, particularly in scenarios with limited measurements, offering a promising avenue for more resilient and efficient power system monitoring and control.

In their study, Revanesh et al. introduced the ANN based Levenberg-Marquardt Neural Network (LMNN) to improve energy efficiency and anomaly detection in Wireless Sensor Networks (WSNs) [11]. By integrating this refined LMNN with protocols like Low-Energy Adaptive Clustering Hierarchy (LEACH) and Energy-Efficient Sensor Routing (EESR), the researchers aimed to extend network lifespan and bolster reliability. The proposed system demonstrated a notable reduction in energy consumption, achieving a 15 percent improvement over traditional methods. Additionally, the neural network-based Intrusion Detection System (IDS) effectively identified anomalies, improving security measures within the network. The simulation results underscored the model's superior performance, marking a significant advancement in the development of intelligent, energy-efficient, and secure WSNs. However, the study also notes that the performance of the proposed model may degrade under high data load or severe network congestion.

Use of deep learning and reinforcement learning

Deep learning (DL) extends artificial neural networks (ANNs) with multiple layers to learn complex patterns, making it highly effective for energy forecasting and grid optimization [12]. Reinforcement learning (RL), on the other hand, enables adaptive decision making through trial and error, with deep RL further improving smart grid automation and real-time energy management. It is increasingly used in industrial automation and optimization, enabling autonomous decision making and adaptive learning from real-world feedback [13]. These machine learning techniques contribute to the security, stability, and efficiency of smart grids and energy systems. In this section, we review significant research works that utilize DL and RL in energy applications.

Smart grid technology improves the dependability, security, and efficiency of electrical grids. However, its reliance on digital communication technology introduces additional risks that must be considered for efficient and trustworthy electricity delivery. H. Karimpour et al. [14] proposed a method for unsupervised anomaly identification based on a statistical correlation between measurements. In large-scale smart grids, the objective is to develop a scalable anomaly detection engine that can differentiate between a true problem, a disturbance, and an intelligent cyber-attack. The suggested method uses feature extraction with symbolic dynamic filtering (SDF) to reduce computing cost while identifying causal relationships between subsystems. The simulation findings on IEEE 39, 118, and 2848 bus systems validate the performance of the proposed technique under various operational scenarios.

The Fourth Industrial Revolution, or Industry 4.0, has emerged with massive amounts of information being. It is the result of the Internet's ability to connect every machine and activity via network sensors. The research work presented in [15] explains the

thinking behind Industry 4.0 and its connection with AI and specifically ML. A comprehensive collection of the most widely used ML and DL algorithms in Industry 4.0 is provided. All algorithms are reviewed and ranked according to their functionality, usefulness, and efficiency. The energy system and its requirements for realizing various applications of Industry 4.0 were discussed and elaborated while forming a connection with AI/ML. Lastly, the developments and challenges in the field of data analysis in the context of the new industrial era, such as scalability, cybersecurity, and big data, are emphasized and examined.

Y. Lu et al. proposed a research study using deep reinforcement learning (DRL) for optimal scheduling of active distribution systems, integrating distributed generation (DG), energy storage systems (ESS), and flexible loads [16]. Their work emphasizes the growing complexity of modern energy systems and the need for advanced optimization methods to handle real-time operational challenges. The proposed DRL-based model efficiently balances supply and demand by optimizing the operation of DG, ESS, and flexible loads, improving system reliability and minimizing operational costs. The proposed model was validated by simulations and demonstrated its effectiveness in improving the scheduling of distributed energy resources. The findings of the study mostly benefited the field of smart grid management, offering a scalable solution for energy systems with high levels of distributed generation and flexibility.

Recently, smart grids have garnered a great deal of interest on a global scale. AI represents the key enabling technology for the smart grid's continued growth and success. The reason is the unpredictability of the smart grid, and the large volume of information that is being collected. AI has entered its new developmental stage, and AI 2.0 is developing at a quick pace due to the rapidly dropping cost of processing power, the proliferation of data, and improvements in algorithm design. Deep learning, reinforcement learning, and its combination-deep reinforcement learning (DRL) are the three methods that are emblematic of the AI 2.0 family and have reached a somewhat mature stage of development. The study presented in [17] introduces the concept and current state of the three aforementioned approaches, and a summary of their potential for application in smart grids is also discussed.

Ting Yang et al. [18] presented a review of RL in renewable energy and electrical systems. Nonlinear systems with multivariate high complexity and uncertainty best describe the dynamic nature of sustainable energy systems, which can vary considerably as the environment and load change. The inclusion of intermittent RES and household energy consumption habits brings more uncertainty into sustainable energy and electric systems. To guarantee the quality of service of sustainable energy and electric systems, their operation, control, and decision-making will necessitate ever-increasing intelligence and flexibility in control and optimization processes. RL is able to develop robust learning abilities and high adaptability because of the dynamic nature of the interactive context. It is believed that RL is well-suited to energy systems that are both sustainable and nonlinear, complex, and uncertain. RL's application to sustainable energy and its related sector is bound to alter the status quo of energy consumption and inject new levels of intelligence into the system. Their research [18] discussed the suitability of RL for sustainable energy and electric systems, explored some current applications of the technique, and looked ahead to some of the challenges and opportunities that lie ahead.

Hybrid models and other AI techniques

Hybrid ML models enhance energy systems by combining different AI techniques for improved prediction and optimization [19]. These models also improve demand response and peer-to-peer energy trading [20]. Hybrid ML models and many other AI techniques are being used in the energy sector to predict energy consumption and to enhance the efficiency of systems. In this section, we review several papers that study the use of those models.

A systematic review of AI and ML approaches to managing energy demand response (DR) is presented in [1]. It is discussed that DR is a cost-effective solution to provide flexibility and improve the reliability of energy networks. Due to the high complexity of the tasks associated with DR, such as frequent and near real-time decision making, efficient management techniques are required, and this is where ML can potentially provide a befitting solution. Methods based on AI can be applied to a variety of problems, including selecting the optimal set of consumers to respond, learning their attributes and preferences, implementing dynamic pricing, and scheduling, and learning how to incentivize and reward participants in DR schemes in a fair and economically efficient manner. Based on a comprehensive review, this study [1] provides an overview of AI methods utilized for DR applications. The studies are categorized based on both the AI and ML algorithms and the energy DR application domain. Next, commercial initiatives (including start-ups and established businesses) and large-scale innovation projects utilizing AI for energy DR are highlighted. The review also notes the lack of real-world deployment data and highlights the need for scalable, adaptive algorithms capable of integrating heterogeneous sources of energy consumption behavior.

In [21], AI is examined for the economic evaluation of energy efficiency and renewable energy technologies. The energy sector is currently confronted with escalating challenges associated with rising demand, efficiency, and lack of analytics necessary for optimal management. The primary factors that incorporate AI are renewable energy technologies, such as energy forecasting, energy efficiency, and energy access. The purpose of their study, was to analyze, compare, and develop a model based on AI that is crucial for renewable energy economic forecasting. AI approaches that can be used to overcome various challenges include selecting the best consumer based on attributes and desires, competitive pricing, scheduling, and facility management; incentivizing demand response participants; and compensating them equally and economically. The proposed model can help increase energy efficiency to 97.32 percent and improve the utilization of RES. However, the study acknowledges that high prediction accuracy heavily depends on consistent and quality datasets, which are still limited across many developing regions.

The study presented in [22] aims to study AI methods for the energy industry. Their research provides a realistic starting point from where researchers and readers can evaluate their own AI projects, goals, state-of-the-art applications, problems, and global policymaking responsibilities. Their discussion is centered on three main points: (i) the use of AI to solar and hydrogen power generation, (ii) the application of AI to the control of supply and demand, and (iii) current developments in AI technology. In this research, the authors looked at different AI application scenarios that can improve manageability, big data processing, cyberattack prevention, and energy efficiency. The results of this research demonstrate that AI is a game-changing technology in the increasingly

competitive energy sector. The paper also outlines that despite numerous pilot implementations, large-scale deployment still faces interoperability challenges due to heterogeneous systems and data silos in legacy energy infrastructure.

The authors in [23] discuss sustainable development as a potential application of AI. AI is rapidly expanding the boundaries of industry, corporate practices, and government legislation. The intelligence of computers and robotics with deep learning (DL) capabilities has had significant disruptive and enabling effects on businesses, governments, and society. In addition, they are impacting broader global environmental trends. More specifically, AI has the potential to increase our pace in achieving Sustainable Development Goals (SDGs) of the United Nations (UN). The researchers combine the viewpoints of business strategy and public policy to analyze the effects of AI on sustainable development, with a particular emphasis on the advancement of the SDGs. Considering managerial learning and leadership development, it also draws some lessons for global sustainability. The authors further note that aligning AI development with sustainability values requires rethinking the reward structures for innovation and integrating ESG (Environmental, Social, and Governance) metrics into AI-driven initiatives.

The study presented in [24] analyzed a significant literature covering various AI algorithms applied in six distinct application sectors. This study aimed to develop a universal procedure to identify viable AI solutions for energy savings. In this study, a hierarchical structure was incorporated into the universal workflow to assist users in selecting learning, optimization, and control tools to achieve energy savings. The established methodology delivers qualitative and quantitative energy-saving effects for use in many fields, based on the findings of numerous investigations. Their work aims to facilitate the commercial application of AI in the field of energy conservation. The universal workflow created in this study provides practical techniques for facilitating the use of AI in a variety of applications. With feasible design standards, commercial uses of AI for energy savings can be anticipated to accelerate.

Smart buildings rely on the Internet of Things (IoT) to enable building operations such as temperature monitoring, safety, and maintenance. As the IoT becomes more pervasive, smart buildings are becoming crucial components of larger system integrations. The IoT plays a significant role in smart buildings and provides amenities that enhance human security through the application of lifesaving technologies. The research work proposed in [25] examines the function of IoT devices in smart buildings. In this review, the IoT device platform and its components are highlighted. In addition, this review identifies security challenges associated with IoT and smart buildings. Various ML and IoT techniques have been identified that are used to improve the energy efficiency and effectiveness of smart buildings. The paper also explores how integrated IoT-ML systems enable predictive maintenance and automated energy management, contributing to up to 17% reductions in HVAC-related energy consumption.

Techniques and algorithms based on AI are increasingly employed to solve a variety of engineering problems. However, the majority of AI research in the energy domain has focused on resolving specific technical problems. There is little discussion on how AI can be utilized to improve the overall operations of the energy system, specifically the electricity market. The study presented in [26] introduces platform architectural logic, which includes both technical and economic considerations for the development of AI-enabled energy platforms for the future electricity market. Using aggregated data from a

European Union (EU) Horizon 2020 project and a Finnish national innovation project, the concept of an AI energy platform is presented. This platform is designed with four integrative layers that could enable distributed energy systems and the electricity market, thereby adding new knowledge and contributing to the existing body of research. The study concludes by discussing the potential applications of the AI-enabled energy platform.

Data from smart meters, sensors, and other IoT devices are just a few examples of the plethora of energy-related data that is being generated by smart buildings. The focus of the work presented in [27] is to introduce a data-driven architecture for implementing real-time processing, data management, and exchange for smart and energy-efficient buildings. This multi-domain big data environment incorporates new AI algorithms and blockchain technology to permit data sharing. The utilization of a wide range of data for the efficient operation of buildings is made possible by data-driven architecture, which also facilitates reliable and effective policymaking and supports the development and deployment of novel energy efficiency services. The study also demonstrates how blockchain integration reduces latency in data exchange by 22% improving the system responsiveness.

The ever-increasing prevalence of converter-interfaced generation and RES makes modern electrical power systems increasingly susceptible to high-impact occurrences, such as extreme weather, which could result in severe disruptions or even blackouts. These contingencies can then propagate to adjacent energy systems via coupling components or technologies, negatively impacting the overall operation and resilience of the multi-energetic system (MES) (such as gas, heating, and electricity). In recent years, ML-based techniques (MLBTs) have been utilized to handle a variety of power system issues, including system planning and security and reliability assessment. The research presented in [28] examines MES resilience measurement approaches and the application of MLBTs to determine the extent of the resilience of future sustainable energy systems.

Due to the growing need for energy in smart microgrids, it is anticipated that peer-to-peer energy trading will comprise a substantial amount of research in the next generation of power systems. However, the on-demand usage of energy is viewed as a significant obstacle to obtaining the optimal household cost. Jamil et al. in [29] offered a blockchain-based predictive energy trading network with real-time assistance, day-ahead control, and generation scheduling. Two modules are proposed for this scheme to work, namely, blockchain-based energy trading and smart contract-enabled predictive analytics. The blockchain module provides peers with real-time energy consumption monitoring, simple control over energy trading, a reward scheme, and immutable energy trading transaction logs. The smart contract-enabled predictive analytics module tries to estimate short-term energy use based on historical energy consumption data. This study uses actual energy usage statistics obtained from the Department of Energy, Jeju Province, Republic of Korea. This research seeks to optimize power flow and energy crowdsourcing to facilitate energy trading between consumers and prosumers. To test the efficacy of the proposed prediction model, they employed several statistical metrics, including mean square error and root mean square error, on a variety of ML models. In addition, the authors analyze the effectiveness of the blockchain platform via Hyperledger Fabric in terms of latency, throughput, and resource consumption. Based

on experimental findings, the suggested paradigm for energy crowdsourcing between prosumer and customer is effective for achieving service quality.

Yao et al. [30] comprehensively reviewed ML-based techniques for a sustainable energy future. The efficient harvesting, storage, conversion, and management of RES is a critical global challenge that must be investigated to meet the goals of the transition from fossil fuels to RES. This transition requires advanced materials, devices, and systems. Researchers in the energy sector have started applying ML strategies to advance these developments. The authors present a set of key performance indicators that can be used to evaluate and contrast the efficacy of various ML-accelerated workflows for use in energy research. In this article, the authors discuss and evaluate the most recent developments in the application of ML to the creation of energy harvesting (photovoltaics), storage (batteries), conversion (electrocatalysis), and management technologies (smart grids).

The paper in [31] presents a framework that uses a 5G-Next Generation-Radio Access Network (5G-NG-RAN) to create a cloud-based digital twin for monitoring wind turbines and predicting power generation. The proposed model addresses the challenges posed by the unpredictability of wind speed, which complicates energy management decisions. Utilizing Microsoft Azure's digital twin infrastructure, the framework employs a deep learning approach that combines a temporal convolution network (TCN) for wind speed prediction and a non-parametric k-nearest neighbor (kNN) regression for estimating power generation over medium-term periods. The model processes univariate time series data to forecast wind speed and power generation for various time frames, from one week to a month. Experimental results on publicly available datasets demonstrate the effectiveness of the proposed framework, showing superior performance compared to classical prediction models. This innovative approach not only enhances remote monitoring capabilities for wind farms but also aids management teams in making informed energy consumption decisions in advance.

The research done by N. Mostafa et al. [32] studies the potential of big data analytics and machine learning in managing renewable energy resources within smart grids. By analyzing vast amounts of data from various sources, these techniques can optimize energy distribution, reduce costs, and improve grid stability. The authors highlight the importance of accurate forecasting, demand response, and anomaly detection in achieving efficient renewable energy integration. Notably, the study emphasizes that integrating ML-driven forecasting with real-time sensor data can reduce grid imbalance events by up to 30%. Overall, the paper provides valuable insights into the role of advanced data analytics in addressing the challenges of renewable energy integration in modern power grids.

Machine learning is not only useful in conventional energy systems, it also can be used in other systems such as waste-to-energy systems. In [33], Zhang et al. present a comprehensive study that investigates the potential of machine learning in optimizing a municipal solid waste-to-energy system for environmental sustainability. The authors aim to enhance the system's efficiency, reduce emissions, and promote hydrogen production. They used different ML models including support vector machines, random forests, and neural networks, to predict system performance and optimize operational parameters. The findings of the research demonstrate the effectiveness of machine learning in

improving the overall sustainability of the waste-to-energy system. This also highlights that ML has potential as a valuable tool for addressing environmental challenges.

Lastly, this section discusses the manufacturing sector which significantly utilizes energy systems. New technologies are required that can lead to sustainable manufacturing. In this regard, intensive research efforts have been made to exploit ML/AI techniques in the manufacturing sector to achieve sustainable production. The study proposed in [34], conducts a systematic literature review on the application of ML in the manufacturing industry. With the advent of Industry 4.0, ML is regarded as the driving force behind the smart factory revolution. The authors suggested that ML-based techniques are the way forward to achieving energy efficiency in the manufacturing process.

Summary of literature on ML in energy systems

A holistic view of the current state of ML applications and approaches for energy systems is presented in this work. To complete the whole picture, energy policy, energy economics, and energy management are also discussed. This integrated perspective helps identify the immediate goals and long-term objectives of ML with respect to energy efficiency, sustainability, and resilience. To provide a structured overview of recent advancements, key studies on energy systems utilizing data analytics and AI/ML are summarized in Table 1, with comparison across their respective focus areas, such as sustainable energy (SE), renewable energy (RE), smart grids (SG), and energy storage (ES).

The table incorporates the most recent review papers that emphasize the growing complexity of deploying ML-based solutions in real-world energy infrastructures. Notably, these works highlight common application areas such as predictive maintenance, demand-side response, and system-level optimization, while also underscoring recurring limitations. These include challenges in data availability and quality, lack of model interpretability, difficulties with integration across legacy systems, and ethical concerns such as bias and privacy risks. For example, studies like [35] and [39] point out the fragmented state of energy datasets and the lack of standardization in evaluation protocols. Meanwhile, [36] and [37] explore critical ethical and privacy implications, reinforcing the need for interdisciplinary approaches and robust governance frameworks. The limitations and challenges identified in the table are further elaborated in Section "[Challenges of implementing ML in energy systems](#)" to provide detailed analysis on the barriers to scalable and trustworthy ML adoption in energy systems.

Table 2 presents a categorization of recent original research works focused on ML-based techniques and applications. The studies are classified based on the ML methods employed, the specific domains they target, such as sustainable energy, renewable energy, smart grids, and energy storage, as well as the key objectives they aim to achieve. It is evident that the sustainable energy sector received the most attention, while renewable energy, smart grids, and energy storage also attracted considerable focus. The quantitative data indicates that the application of ML has led to notable performance improvements. However, due to significant differences in the nature and scope of ML applications across these studies, direct quantitative comparison of techniques is not feasible. These methodological differences include variations in model architectures, training datasets, evaluation metrics, and deployment environments. Deep learning approaches have shown strong potential to improve the system stability. Reinforcement

Table 1 Summary of the recent and relevant review papers focused on ML-based techniques in the energy sector

Ref	Year	AI Technique	Focus				Significance	Limitations/challenges
			SE ¹	RE ²	SG ³	ES ⁴		
[1]	2020	DL	✓				Demand-side Response	Generalizability & adaptation challenges
[2]	2021	RL	✓	✓	✓	✓	Energy Efficiency Enhancements in Smart Grids	Scalability, sparse reward signals, & convergence issues
[15]	2021	AI, ML, and DL	✓		✓		A new architecture for AIA is presented	Lack of standardization & integration difficulty
[22]	2021	AI		✓	✓	✓	Big data, cyberattack prevention, smart grid, IoT, energy efficiency	Challenges in data interoperability, security, and trust in AI decisions
[34]	2020	AI and ML			✓		AI and ML applications in smart production and industry	High computing costs & limited deployment
[35]	2025	SVM, KNN, DL	✓		✓		Power quality disturbance classification using AI	Dataset scarcity, synthetic data reliance, limited real-world generalization
[36]	2024	ML	✓	✓			Socio-technical framework for energy justice in ML models	Bias in training data & need for interdisciplinary collaboration
[37]	2024	ML		✓	✓		Ethical risks in NILM and smart metering	Autonomy loss & unequal access
[38]	2024	ML		✓	✓		Secure data handling in large-scale smart meter networks	Data volume, governance issues, secure and scalable storage gaps
[39]	2024	ML, AI	✓	✓	✓	✓	Barriers to AI integration in modern power systems	Fragmented data, poor real-time processing, regulatory gaps

¹ Sustainable energy
² Renewable energy
³ Smart grid
⁴ Energy storage

learning techniques are particularly effective in reducing operational costs. In addition, blockchain-based methods have also demonstrated significant improvements.

Research design

This chapter presents the systematic methodology employed in this review, providing a comprehensive analysis of recent advancements, existing challenges, and future directions in machine learning (ML) applications for energy systems.

A rigorous systematic review methodology was adopted for this manuscript. The review process comprised several critical stages. A targeted search strategy was employed to identify relevant literature using Boolean operators and predefined keywords across multiple academic databases, including IEEE Xplore, Elsevier ScienceDirect, SpringerLink, ACM Digital Library, MDPI, Scopus, and Google Scholar. The study considered peer-reviewed articles and conference papers published between 2019 and 2025 to ensure coverage of recent developments.

A strict inclusion and exclusion criterion was applied to maintain the quality and relevance of the selected studies. Only publications explicitly addressing ML applications in energy systems from the past five years were considered, while studies unrelated to ML-driven energy solutions, non-English publications, and those lacking empirical validation were excluded.

The screening and selection process followed a three-stage approach: initial study identification through database searches, followed by title and abstract screening to remove irrelevant works, and finally, a full-text review to confirm adherence to inclusion criteria. The search queries were meticulously designed to encompass a broad spectrum of research themes related to ML in energy systems.

For data extraction, a structured framework was implemented to systematically capture critical information from the selected studies. Key aspects such as ML methodologies, application domains, datasets, evaluation metrics, and identified challenges were documented. This structured approach ensured consistency and comparability across the reviewed literature, facilitating a comprehensive synthesis of the findings. The screening process and the selection criteria are summarized in Fig. 2.

Applications of machine learning in the energy sector

The growing complexity and scale of modern energy systems demand intelligent, data-driven solutions that can adapt to uncertainty, enable real-time decision making, and enhance operational efficiency. Machine learning (ML) has emerged as a key enabler in this transformation, offering robust tools to analyze vast energy datasets, identify patterns, and support automation across various domains. This section highlights how ML is being applied practically in the energy sector, focusing on real-world implementations and research-aligned use cases in energy distribution, storage, forecasting, demand-side management, smart grid operations, and energy efficiency. The goal is to illustrate the breadth of ML's operational impact, ranging from infrastructure-level optimization to intelligent control of decentralized systems.

ML applies mathematical and statistical techniques to reveal meaningful information [40]. It concentrates on programs that take advantage of prior experiences to improve prediction and decision-making abilities. The steps involved in developing an ML-based solution are illustrated in Fig. 3.

Table 2 Summary of the recent research works focused on utilizing ML-based techniques

Ref	Year	ML technique	Focus		Significance			Results
			SE ¹	RE ²	SG ³	ES ⁴		
[6]	2025	ML	✓			✓	Optimize energy harvesting efficiency in IoT networks	85% improvement in energy efficiency
[7]	2023	ANN	✓				Energy production and monitoring	15% efficiency improvement
[8]	2023	ANN	✓				Prediction of Thermal Properties	Thermal Conductivity MAE = 0.04 W/mK
[12]	2025	DL	✓			✓	Optimize voltage and frequency control	Improved stability by 37%
[13]	2025	ML	✓	✓			Improve predictive maintenance & quality control	20% downtime reduction
[18]	2020	RL	✓	✓			Energy management, grid optimization, renewable integration	Reduced operational costs by 18%
[19]	2025	Hybrid				✓	Optimize green building energy efficiency	25% energy savings in simulation
[21]	2021	AI	✓	✓			AI for economic evaluation	97.32% increase in energy efficiency
[23]	2020	AI	✓				Improved economic, environmental, and social sustainability	Life Cycle Assessment improved by 16%
[24]	2021	ML		✓		✓	Universal workflow	High scalability
[25]	2022	ML		✓			Internet of Things, smart buildings	17% reduction in HVAC energy use
[27]	2020	Blockchain		✓			Smart meters' data, sensors and other IoT	Latency reduced by 22%
[28]	2022	ML	✓	✓			Multi-energetic system (MES)	14.2% efficiency increase
[29]	2021	Blockchain				✓	Blockchain-based predictive energy	Improved fault detection by 29%
[30]	2023	ML				✓	Energy harvesting, storage, conversion, and management	Efficiency improved by 23.1%

¹ Sustainable energy

² Renewable energy

³ Smart grid

⁴ Energy storage

Figure 4 shows common ML models currently deployed in smart energy systems. These models, including supervised, unsupervised, and reinforcement learning algorithms, are applied to real-time load forecasting, energy efficiency improvement, anomaly detection, equipment maintenance, and renewable integration.

How is machine learning transforming the distribution of energy?

ML is changing energy distribution by enabling more efficient, cost-effective, and reliable methods for managing and optimizing energy systems. ML algorithms can be used to analyze large volumes of data from various energy sources, such as wind, solar, and fossil fuels, to predict energy demand and optimize energy distribution in real-time. This helps energy providers make informed decisions on energy generation and distribution and, more importantly reduce energy waste as well.

The utilization of powerful ML predictive models along with data analytics have benefited renewable energy utilities greatly [22]. Predictive models have helped reduce expenses, improve forecasts, and maximize portfolio returns. ML models perform various tasks in energy applications, such as storing data [41], visualizing data [42], analyzing data [43], capturing data [44], updating data, etc. The growing interest in ML can be attributed to several factors, including better computational abilities, and advanced ML techniques [45]. ML can assist the energy industry in various ways, such as in grid management [46], demand response [47], maintenance [48], energy source exploration [49], reducing electricity use to save costs [50], and so on.

Energy sector exploiting machine learning: practical implementations

In this section, we provide a brief discussion on how the prominent names in the energy industry are utilizing ML for targeting various applications.

Optimization of yield: General Electric's renewable energy division uses ML to create a cloud-based platform that simulates a wind farm. This virtual model is based on real wind patterns and predicts energy output to optimize performance on either an individual turbine or farm-wide level. This approach is projected to result in a 20% increase in electricity efficiency and save up to \$100 million over the lifespan of a 100 MW wind farm [51].

Energy forecasting: The Vermont Electric Power Company in the US is utilizing data science and ML techniques to develop hyper-local weather prediction systems. This application of ML has resulted in a significant reduction in solar energy and wind energy prediction error. This has led to cost savings of \$1 million per 1% reduction in load through improved resource and operational management [52].

Energy storage: Greensmith Energy, a multinational energy storage company, uses ML to control energy storage devices and environments. It helps RE-based power plants to deliver power local grid, reducing the reliance on fossil fuels and cutting down greenhouse gas emissions [53].

Grid behavior interface: One of the leading Swiss Energy companies, "Gridsense" uses ML to gather information on grid requirements, energy demand, environmental prediction, and electricity pricing. This information is critical for load balancing, reducing power grid peak loads, and stabilizing power delivery [54].

Energy trading: Typically, ML and automation together can be leveraged for energy trading. The automation performs repetitive tasks, freeing up analysts to focus on more

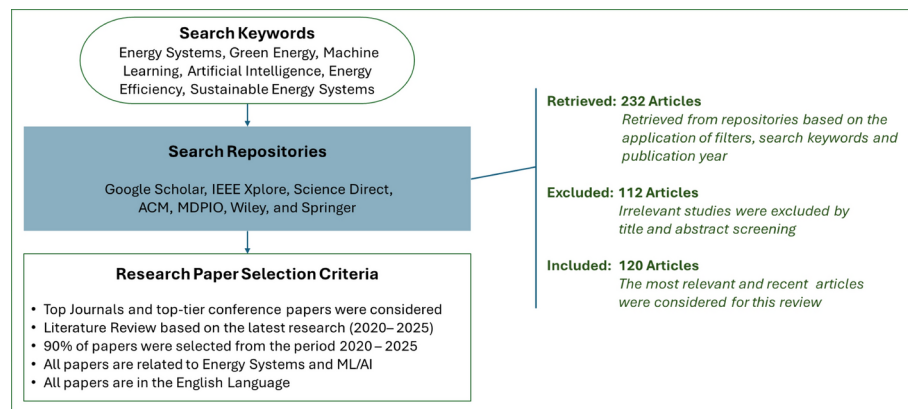


Fig. 2 Research article selection criteria

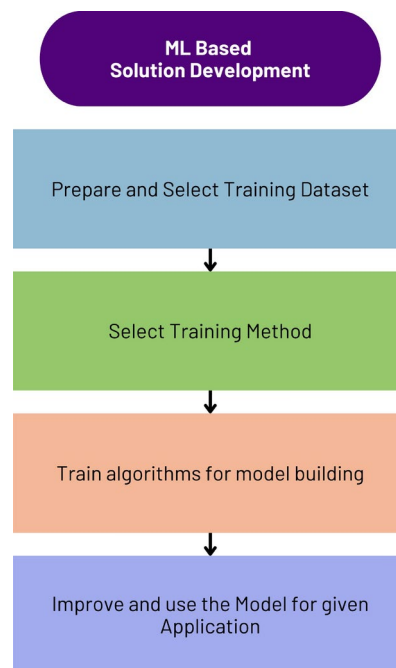


Fig. 3 Steps to develop ML-based solutions

critical activities such as analyzing and contextualizing data. The data gathered can then be used by ML to provide more useful insights. British Petroleum utilizes ML for energy trading [55].

Complaints management: A US-based power and gas utility company, Exelon, created an ML-operated chat-bot to address customer issues like billing and outages. This led to a reduction in the number of complaints and greater insight into customer consumption needs [56].

ML is increasingly being used in core energy technologies to improve various aspects of energy production, distribution, and consumption. Some examples include:

1. Predictive maintenance in power plants to improve equipment efficiency and reduce downtime.

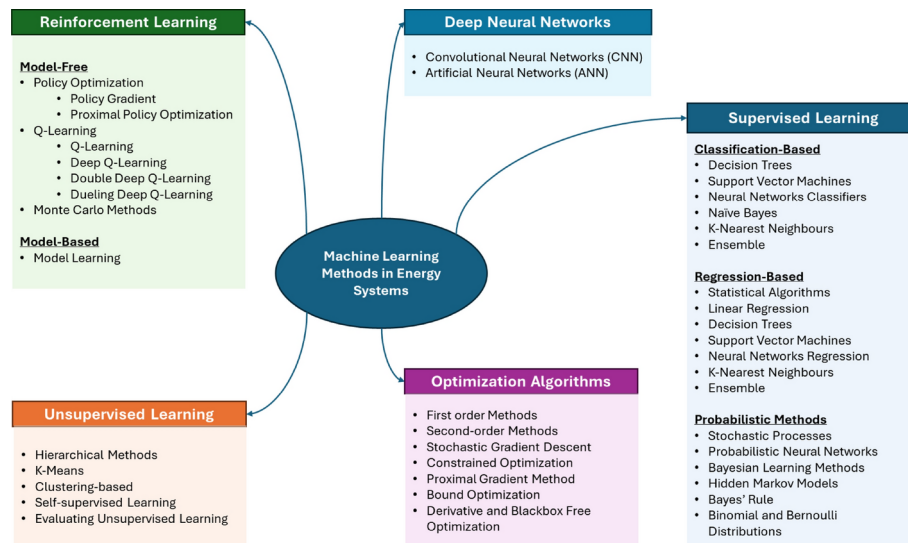


Fig. 4 Various machine learning models used in smart energy systems

Table 3 Real-world applications of machine learning in the energy sector

Company/domain	Application area	ML contribution	Impact/result	Ref
General Electric (GE)	Yield Optimization	Simulates wind farm performance	20% gain, \$100M savings	[51]
Vermont Electric Power	Energy Forecasting	Forecasts RE using weather data	\$1M saved per 1% load accuracy	[52]
Greensmith Energy	Storage Control	Manages storage and grid delivery	Reduced fossil use and emissions	[53]
Gridsense (Switzerland)	Grid Interface	Analyzes grid demand and pricing	Better load balance and stability	[54]
British Petroleum (BP)	Energy Trading	Automates insight and decisions	Faster, smarter trading responses	[55]
Exelon	Customer Service	ML chatbot for billing issues	Fewer complaints, more insights	[56]

2. Optimizing energy production through ML algorithms that predict energy output based on weather patterns.
3. Improving energy efficiency in buildings through smart energy management systems that use ML to predict and control energy usage.
4. Automating the detection and repair of power grid failures using ML algorithms.

Table 3 summarizes the diverse real-world applications of machine learning across the energy sector, showing its tangible impact on performance, cost savings, and system reliability. From optimizing wind farm output and improving load forecasting to managing energy storage and automating grid operations, ML enables smarter, data-driven decisions across the energy value chain. It also supports non-technical functions like energy trading and customer service, showing its broad applicability.

Fundamental energy technologies and machine learning

This section presents the impact of ML on the fundamental energy technologies necessary for planning, designing and maintaining energy systems.

Advanced energy materials research

ML has the potential to bring about new breakthroughs and methods of producing energy efficient materials [57]. Recent research shows that data-driven materials engineering is leading to better design and production of advanced energy materials. Machine learning is increasingly being used to accelerate the discovery and optimization of materials for batteries, fuel cells, solar cells, and thermoelectrics by predicting material properties, guiding high-throughput simulations, and identifying promising candidates from large chemical and structural spaces [58, 59].

Computational techniques have advanced that aid research related to materials structures and their electronic properties. For instance, supercomputers are vital for simulating material characteristics and predicting properties. Energy material R&D spending by OECD countries has increased from less than \$10 billion in 2000 to over \$16.6 billion in 2016 [60]. However, the potential for transforming data into practical applications remains untapped [60]. ML algorithms can speed up energy material discoveries but there are challenges, such as the difficulty in describing materials universally, limited availability of scientific evidence, and unreliable predictions and simulations.

To realize the full potential of ML in advanced energy materials, the energy and material sciences need to collaborate with ML communities. Governments must provide adequate funding to support this partnership and their mutual commitments. The use of ML in materials science is expected to see a significant increase in investment globally. An interdisciplinary approach will facilitate the resolution of technical challenges more efficiently [61].

Energy storage technologies

A wide range of energy storage technologies have been developed to meet the energy demands of the utility grid. These technologies are summarized in Fig. 5 [62–65].

The researchers believe that as technology progresses, better energy storage systems and devices with improved capacity, efficiency, and lifespan will be developed [66]. The advanced control methods rely on the real-time status of certain parameters, requiring a balance between many factors in their development. These advancements rely heavily on sophisticated control methods that require real-time monitoring and balancing of multiple system parameters. As the complexity and scale of energy storage systems grow, the role of machine learning (ML) in their development becomes increasingly vital [67].

Recent studies highlight the growing application of ML in the discovery and performance prediction of energy storage materials. Shen et al. [68] provide an extensive review of ML's role in revolutionizing materials research, particularly for dielectric capacitors and lithium-ion batteries. Similarly, Huang et al. [69] emphasize that, despite widespread use of decades-old material systems in mainstream batteries, ML is now enabling faster and more accurate exploration of new materials. ML models are increasingly being used to predict material performance, identify structure-property relationships, and guide experimental efforts, reducing the time and cost traditionally associated with trial-and-error methods.

Energy efficiency

Renewable energy agencies estimate that efficient energy systems and green technologies can reduce up to 90 percent of energy-related CO₂ emissions [53]. As a result,

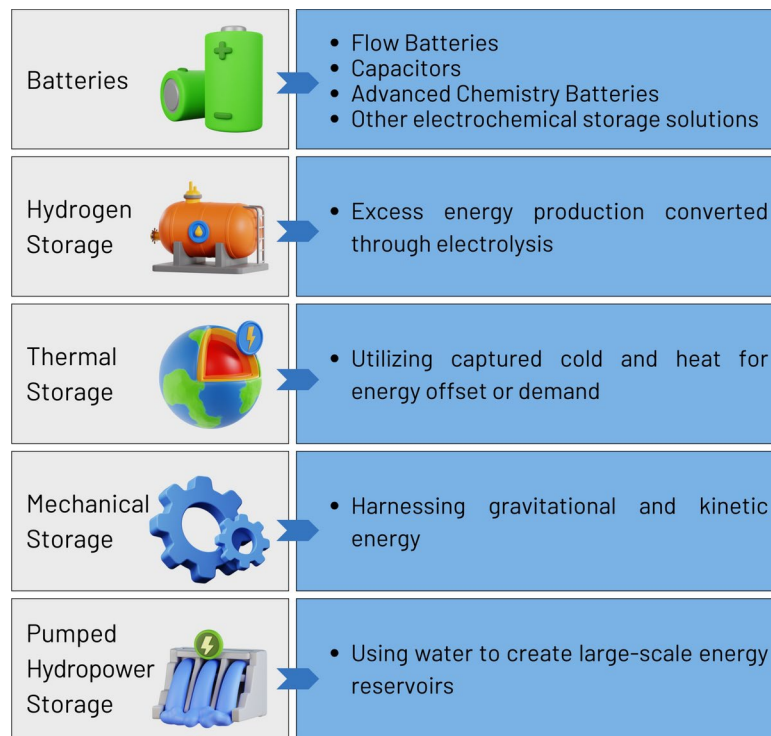


Fig. 5 Energy storage technologies

scientists, researchers, and engineers are placing greater emphasis on improving the energy efficiency of systems across sectors such as buildings, industry, and power grids. Since energy consumption affects everyone, a unified and intelligent approach is essential for achieving optimal management.

Machine learning is increasingly applied to support energy efficiency by enabling smarter system design and the development of strategies that reduce energy waste and improve usage patterns. In smart buildings, machine learning algorithms are used to optimize heating, ventilation, and air conditioning systems, lighting schedules, and appliance operations while maintaining user comfort [70]. In industrial applications, machine learning assists in identifying process inefficiencies and supports energy-aware scheduling to reduce unnecessary consumption [71]. At the grid level, it contributes to load forecasting, demand response, and real-time balancing of electricity supply and demand [53].

Two commonly used approaches for improving energy efficiency include simulation based optimization methods [72] and surrogate based optimization methods [71]. Although these methods are effective, they often involve high computational costs when searching for near-optimal solutions. Machine learning enhances both approaches by reducing computation time and accelerating the discovery of efficient configurations.

In line with the global 2050 target to reduce carbon emissions from industrial processes, machine learning is being used to improve the performance of existing energy systems and support the integration of renewable energy. Its growing role in smart buildings, energy infrastructure, and transport systems reflects its importance in advancing sustainable and intelligent energy solutions.

Demand side management

ML can be used to manage energy demand and influence market response by enabling consumers to interact with and adapt to real-world market dynamics through intelligent technologies [73]. In smart homes, digital technologies powered by machine learning allow for effective energy demand management by connecting devices such as energy storage systems, solar panels, smart meters, and household appliances to the Internet for data collection and real-time decision-making [74]. The demand-side management consists of four key components, including i) enabling technologies such as RES for residential heating, in-home batteries, ML applications, and electric vehicles for smart charging; ii) market design, including net billing and incentive-based tariff schemes; and iii) system operations, including the production of renewable energy and advanced prediction of parameters.

Smart manufacturing in the smart grid

The foundation of the ML structure starts with collecting data from various devices. Industry 4.0 connects machines and operations to the Internet via network sensors which produces vast amounts of data. ML is applied to process and analyze this data to optimize manufacturing operations, improve efficiency, and enable predictive maintenance, making the entire system smarter.

Information from manufacturing equipment, output metrics, and logistics is crucial for building a robust industrial ML framework. The use of hardware components such as sensors, actuators, and embedded ML units must be considered when acquiring data. ML models can then be trained on this data to detect anomalies, forecast equipment failures, and streamline workflows. The next important task is to form coordination between industry sectors to ensure efficient data collection while also securing the data.

To accomplish all the tasks mentioned above, a six-tiered pyramid for industrial ML architecture is presented in [75]. The first layer encompasses the industrial sector, while the second layer caters to the demands of Industry 4.0. The third layer highlights potential challenges faced by ML systems. The fourth layer focuses on constructing industrial ML systems through the use of enabling technologies. The fifth layer involves programming techniques, deep learning or ML methods, and platforms utilized in building the framework. Finally, the sixth layer refers to the potential impact of an AI framework. A detailed study on ML in advanced manufacturing industries has been carried out by T. Kotsiopoulos et al. [15].

Energy planning

Energy forecasting, management, and operation have long been a focus of decision-makers, particularly regarding energy demand and resource allocation. Researchers have applied various decision-making techniques to help choose optimal options, create energy strategies, and provide alternative energy solutions in uncertain situations [76]. Strategic energy planning utilizes ML models to enhance understanding of the market needs for integration into basic and advanced energy infrastructure, and to reach agreement on the best integration strategies and industry practices focused on these strategies [77].

The integration of IoT into energy management is the most significant advancement in ML in the energy industry. IoT provides remote access, automated programs, data

analysis, and intelligent tracking through the use of smart meters and sensors [78]. These devices inform consumers of energy consumption in both manufacturing processes and equipment. Although utilities recognize the significant impact ML could have on their businesses, they have yet to incorporate it into their key strategies.

The energy industry faces significant challenges due to uncertainty in the modern energy landscape [79]. Conventional market models, which feature steady control, complete integration, ensured long-term ROI (Return on investment), and consistent power generation, are struggling to keep pace and are being replaced by more affordable, self-sufficient, and digitized ML innovations, creating new business opportunities [80]. The modern energy landscape's uncertainty presents significant challenges for both utilities and independent power producers. The traditional energy market models are being outstripped by new, affordable, autonomous, and digitized ML innovations, which will bring new business opportunities.

Renewable energy integration

According to a survey, to address global warming, renewable energy production needs to grow at an eight times faster pace, with \$131 trillion in investment required over three decades. The International Renewable Energy Agency (IREA) believes that this requires immediate action and an investment of \$131 trillion in renewables by 2050. In recent decades, two-thirds of the energy consumed was from fossil fuels, but this will be limited to 10 percent by 2050 [53]. ML is applied to address the challenges posed by the intermittent nature of renewable energy sources, particularly in maintaining grid stability during the low-carbon transition. Key applications of ML in renewable energy integration include identifying barriers to green energy adoption, enhancing market forecasting, and supporting more effective resource management.

The energy sector faces pressing global challenges that require immediate attention. As policymakers increase renewable energy deployment and reduce reliance on fossil fuels, the energy market is expected to undergo significant disruptions. To manage this transition and align with the goals of industry 4.0, grid operators, engineers, and consumers are applying ML techniques [81]. ML is used for advanced prediction of energy supply and demand, asset condition monitoring, and real-time market forecasting, allowing for more efficient system operation and planning [53].

Smart grid

The large number of customers, entities and operational activities in the energy systems generate thousands of terabytes of data each year. To keep pace with this data growth, power utilities must store and analyze large amounts of energy and climate data [82]. Therefore, ML-based approaches along with big data analytics is considered vital for SGs. A global survey of industry experts and electric utilities from ten countries found that big data analytics are critical for smart grids and a potential market opportunity for 80% of electric utilities [83].

ML assists power utilities in managing big data through various stages, including collection, preservation, and retrieval of data from various sources [84]. Data analysis can encompass various forms of analytics, such as descriptive, predictive, informative, and prescriptive, based on the intended usage. Descriptive analytics are commonly used to describe grid and consumer behavior, while diagnostic models evaluate operating

conditions and grid operator decisions. The diagnostic approach aims to identify the causes of a problem and is suitable for proactive action. Predictive analytics are meant to anticipate operational scenarios and aid the decision-making process, as the primary goal of data analysis is to adopt a preventive approach.

However, big data also presents significant challenges for ML and data analytics, such as variations in data formats, reliability of predictive analytics, dispersed input sources, high dimensionality, poor data quality, imbalanced energy consumption, and unclassified and unsupervised data. Despite these challenges, big data enables ML algorithms to extract insights and generate more precise and timely forecasts. The scalability of models and distributed computation are among the major obstacles to ML in big data environments [85].

Machine learning for renewable energy systems

Renewable Energy (RE) has gained tremendous attention as we move towards green and sustainable energy systems. RE has numerous types such as solar, wind, hydro, bio-fuels, geothermal, etc. Among these, solar energy is widely considered in various parts of the world. Figure 6 describes global energy consumption of the year 2022 and the global renewable energy generation of that year [86].

In all these RES, nonlinearity is a common feature. In a hybrid grid-connected system, the integration of these RES can lead to unexpected and abrupt fluctuations, which deteriorate the power generation process. It significantly affects the stability and reliability of grid-connected hybrid systems. To counter this effect, there is a need to predict the behavior of solar and wind sources which will help to formulate better strategies, maintaining the stability of the grid.

In a grid system connected with solar PV plants, the variable and non-controllable nature of solar irradiation production poses critical problems to the performance of the power system. Voltage fluctuation, reactive power compensation, harmonics, frequency response, reliability, and stability are well-known problems. Therefore, accurate forecasting of the PV system's output power is an essential task to address these problems.

ML provides a befitting approach to the problems of RES as discussed above. However, data collection, collection frequency, and optimizing existing algorithms for energy

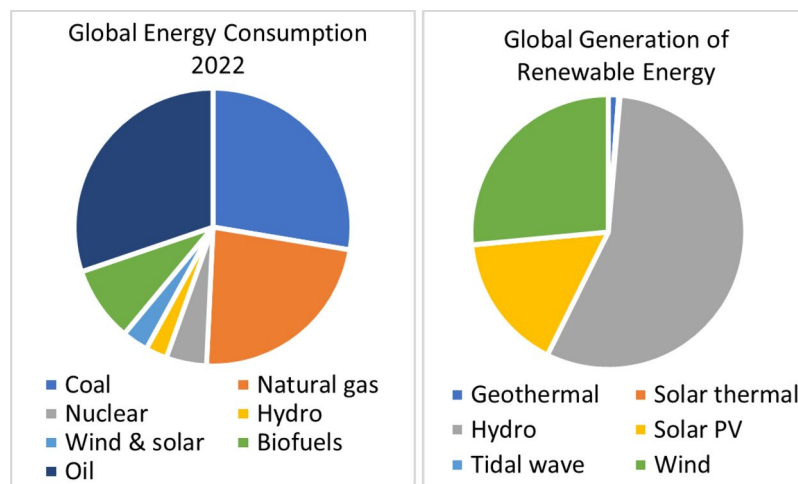


Fig. 6 (A) Global energy consumption 2022. (B) Global generation of renewable energy

systems still need to be investigated. The forecasting accuracy of the trained and generalized model also depends on its forecast horizon. It is the time span over which prediction is performed. There are four types of forecasting horizons: very short term (1 sec– 1 hour), short term (1 hour– 24 hours), medium term (1 week– 1 month), and long term (1 month– 1 year). It is evident from the literature that forecasting accuracy decreases with the increase of horizon [87].

Conventional mathematical techniques such as Persistence and Statistical techniques have been applied to forecast the power generated by PV systems. Unfortunately, these techniques generally produce low forecasting accuracy and fail to adapt to nonlinear data. Due to these limitations, ML techniques such as SVM, ANN, Extreme Learning Machine (ELM), DL, and metaheuristic techniques are used to deal with nonlinear data and enhance the forecasting accuracy [88–92].

Among ML techniques, ANN in [93–96] and adaptive neuro-fuzzy inference system ANFIS in [97] are deployed for solar output power forecasting. They have a better ability to deal with nonlinearity and cope with the variable behavior of solar power. However, the issues of random initial data, local minima, over-fitting, and increased complexity due to multi-layered structure affect the reliability of the power system [98–100]. The SVM has also shown promising results to predict solar power [101–103]. However, it is highly sensitive to parameters such as kernel function, tube radius (ε), and penalty factor (C). Therefore, the proper selection of these parameters is a challenging task [88].

Convolutional neural network (CNN) [104] and Recurrent neural network (RNN) [105, 106] are also used for forecasting of PV power. These deep learning models help to extract features from PV output power and provide better forecasting results than persistence, physical and statistical models. Figure 7 summarized different ML-based forecasting techniques utilized for different RES.

It is interesting to observe that while other fields utilizing DL have matured, there is still room for improvement in selection of optimal hyper-parameters utilized for

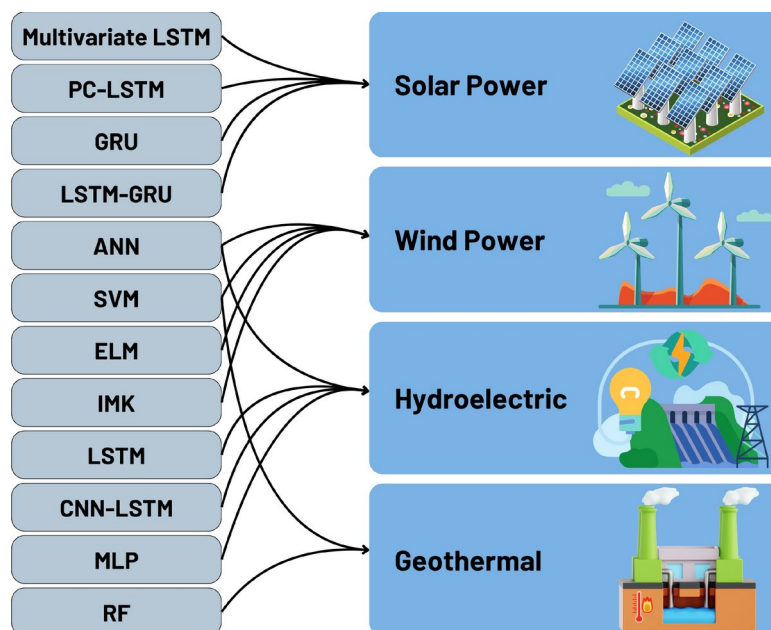


Fig. 7 ML-based forecasting techniques for different renewable energy resources

training on energy datasets. Therefore, the potential investigation area would be to explore advanced optimization algorithms in conjunction with deep learning. Some examples of these algorithms are Particle Swarm Optimization (PSO), Salp-Swarm Algorithm (SSA), and Game Theory (GTO) optimization. The future demands the use of efficient algorithms (faster convergence speeds) that can provide better forecasting accuracy for various energy systems.

Table 4 presents a focused comparison of popular machine learning techniques used in renewable energy forecasting, emphasizing model-specific characteristics that are often overlooked in broader discussions. This table presents strength and practical trade-offs of various ML models in terms of convergence speed, parameter sensitivity, and computational requirements. For example, although ELM is not as widely used as ANN or SVM, its ability to achieve fast convergence with relatively low MAE demonstrates its suitability for very short-term applications where computational efficiency is critical. In contrast, models such as CNN and RNN show strength in capturing complex temporal patterns, but their effectiveness heavily depends on the availability of large datasets and stable training conditions. Although accurate, ANTIS illustrates how hybrid methods can incur significant computational overhead due to structural complexity. This table provides a clear benchmark that supports method selection based on specific forecasting needs, rather than relying on general-purpose popularity.

Challenges of implementing ML in energy systems

Although machine learning (ML) presents transformative opportunities for energy systems, there are several challenges to implement it in real-world. These challenges should be addressed to ensure the effective and sustainable deployment of ML-based solutions in energy applications. This section highlights the main concerns that limit the full potential of ML in energy systems.

Data quality and availability

As mentioned in Section "Smart grid", the effectiveness of ML models in energy systems is highly dependent on high-quality datasets. The integration of artificial intelligence (AI) in energy systems, particularly for stability monitoring and control, is highly dependent on the quality of input data. However, significant challenges arise due to data quality issues in synchrophasor applications, which are critical for real-time monitoring in power grids [107]. The major challenges are synchronization errors, data loss,

Table 4 Performance comparison of ML techniques for renewable energy forecasting

Model	Forecasting Horizon	Accuracy/MAE	Strengths	Limitations	Reference
ANN	Short to Medium Term	MAE ~5–7% (dataset dependent)	Good at modeling nonlinearity	Prone to overfitting, local minima	[93, 94]
SVM	Short Term	High accuracy, sensitive to parameters	Robust to high-dimensional data	Requires careful tuning (C , ϵ , kernel)	[101, 103]
ELM	Very Short to Short Term	Fast convergence, MAE ~6%	Simple, fast training	Lower generalization ability	[88]
CNN	Short Term	Better than statistical models	Effective feature extraction	Needs large datasets	[104]
RNN	Short to Medium Term	Improved temporal learning	Captures temporal dependencies	Training instability	[105, 106]
ANFIS	Short to Medium Term	High accuracy, but computationally expensive	Fuzzy logic integration	Increased model complexity	[97]

and latency, all of which can compromise the reliability of AI-driven decision making. The researchers in [108] explore how Phasor Measurement Units (PMUs) provide high-resolution data for machine learning (ML)-based stability analysis. They focused on the necessity of robust data acquisition and processing techniques to mitigate data reliability concerns. The effectiveness of AI in Wide-Area Monitoring Systems (WAMS) depends on the accuracy and consistency of the input data, as errors in measurement can lead to flawed stability assessments and delayed responses to critical contingencies.

Challenges of data quality are also addressed in [35], where the authors highlight that although AI models, such as SVMs, KNN, and deep learning techniques, have demonstrated high accuracy in power quality (PQ) disturbance detection and classification, their practical effectiveness is often limited by the scarcity and inconsistency of high-quality, real-world datasets. This data inadequacy comes from a combination of limited access to real operational data, labeling difficulties, and the underrepresentation of rare yet critical PQ events. As a result, many models are trained and validated on synthetic or simulated data, which may not capture the full spectrum of variability and noise present in live grid conditions.

All of those studies emphasize the urgent need for standardized data quality metrics and advanced pre-processing methodologies to improve AI applications in power system stability. Addressing these challenges is essential to ensure that AI-driven monitoring and control systems operate with high precision, thereby improving the overall security and efficiency of modern energy grids.

Ethical and policy concerns

The adoption of ML in energy systems is also influenced by ethical and regulatory policy concerns. Biases in ML training data can lead to unfair decision-making and suboptimal energy distribution strategies [109]. Chen et al. address the implications of advancing energy justice by AI bias [36]. The authors dissect the ML processes to reveal key points where bias can be introduced. The potential areas where bias may arise in energy systems are during data collection, model development, and deployment. Such biases may arise from uneven access to infrastructure, historical disparities in energy consumption data, or lack of representation of marginalized communities in training datasets. Their analysis also reveals that addressing ML bias is not merely a technical challenge but a socio-technical imperative that must be tackled through interdisciplinary collaboration, inclusive data practices, and proactive regulatory oversight.

Another important ethical challenge regarding the use of ML in energy systems is the data privacy issues. Gavornik et al. [37] highlight privacy and security concerns in smart metering and NILM, as well as ethical risks such as loss of autonomy, algorithmic bias, and unequal access to infrastructure. These risks are amplified by the granularity of smart meter data, which can expose sensitive user behaviors. Hseiki et al. [110] emphasize the need for robust cybersecurity in smart energy meters, proposing a multi-level architecture using LoRaWAN and unidirectional data flows to counter threats such as DDoS attacks and data tampering. Yadav et al. [38] further identify data management challenges in large-scale smart meter deployments, including high data volume, secure storage, and governance. These studies emphasized the necessity of embedding privacy, security, and ethical stewardship at the core of AI-driven energy systems to ensure resilience and public trust.

Other notable challenges

In addition to the challenges mentioned earlier, several sector-specific concerns significantly impact the effective implementation of machine learning (ML) in energy systems. They include, but are not limited to, computational complexity and scalability, forecasting accuracy and uncertainty, integration with existing energy systems, and reliability of predictive maintenance systems [111, 112].

Moreover, Shobeiry [39] identifies a range of critical obstacles that limit AI integration into modern power systems, including fragmented data management, poor data quality, limited real-time processing, grid coordination complexity, cybersecurity risks, and inadequate regulatory frameworks. Allal et al. [113] further highlight key limitations in ML-enhanced renewable energy systems (RES), such as reliance on scarce materials, inconsistent data across platforms, and challenges in validating models under dynamic, real-world conditions. They also stress the lack of standardization and transparency in training and testing processes, which obstruct reproducibility and scalability. These studies reveal that despite the promise of ML, unresolved issues around data integrity, system interoperability, model evaluation, and regulatory support continue to impede widespread adoption in the energy sector.

Recommendations and future research directions

Machine learning (ML) continues to reshape energy systems, offering advanced capabilities for optimization, forecasting, and decision-making. However, despite the progress made, several challenges must be addressed to fully harness its potential. The successful integration of renewable energy sources, the need for more accurate forecasting models, and large-scale optimization of energy systems remain key areas that require further investigation. Additionally, ensuring high-quality datasets, improving predictive maintenance strategies, and exploring sustainability measures are crucial for enhancing the reliability and efficiency of ML-driven energy solutions.

This section outlines key recommendations and future research directions to bridge existing gaps and push the boundaries of ML applications in energy systems. It highlights the need for improved forecasting techniques, the exploration of alternative ML models beyond deep learning, and the development of better energy datasets. Furthermore, it discusses the role of ML in predictive maintenance, optimization, sustainability, and policy-making. Addressing these areas will not only enhance the performance of energy systems but also contribute to their long-term stability, efficiency, and sustainability.

Renewable energy integration

There are significant advantages to RE due to which it is considered the most promising means of electricity generation. RE is green and clean but it comes with scheduling issues that pose a threat to the stability of the energy system [114]. This problem becomes significant when large-scale integration of renewable energy takes place. The randomness of renewable energy generation is also a challenge as well. Therefore, accurate renewable energy forecasting is of utmost importance. However, to bring accuracy to this prediction, a significant amount of work is still required. Various techniques have been reported in the literature that are commonly used for renewable energy prediction. These techniques are summarized in Fig. 8.

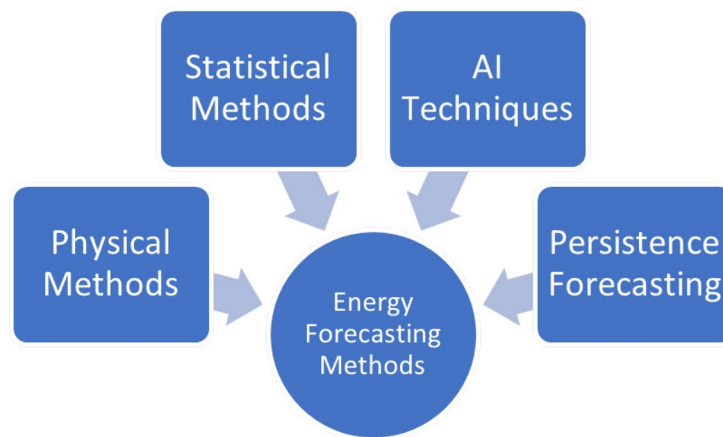


Fig. 8 Commonly used energy forecasting methods

Currently, more emphasis is given to the first three techniques whereas the requirement is to focus more on AI techniques, specifically ML algorithms. The current forecasting techniques are predominantly based on deterministic prediction whereas the deep learning-based probabilistic techniques need to be investigated. Deep learning can help identify and quantify the uncertainties in renewable energy generation. This will not only help with accurate forecasting and bringing stability and reliability to the energy systems but also has economic significance. Moreover, the distribution and management of energy system becomes streamlined as well.

Future work should focus on developing hybrid ML models that integrate physical system knowledge with data-driven approaches, especially for short-term and ultra-short-term forecasting in volatile climates. Additionally, benchmarking these models using standardized datasets from diverse geographic regions is necessary to assess generalizability.

It has been observed in the literature that the ML algorithms are mostly applied to specific renewable energy types, typically wind and solar. However, it should be extended to other renewable energies as well. Accurate prediction of other resources will help in maintaining the capacity of energy systems and serve the goal of large-scale integration of RE to the existing energy systems. Specifically, biomass, geothermal, and tidal energy systems remain underexplored in ML literature and present viable research opportunities. Future studies should aim to develop forecasting tools tailored for these sources, considering their unique temporal and spatial characteristics. Furthermore, research should also investigate the operational and economic implications of integrating these underrepresented RE types, and explore ML strategies capable of capturing their nonlinear and site-specific generation patterns.

Exploiting other-than-deep learning models

While deep learning (DL) models have shown considerable promise in a range of energy system applications, there remains significant untapped potential in other machine learning (ML) models. A review of the recent literature reveals that DL continues to dominate the methodological landscape, often at the expense of classical or alternative ML models such as support vector machines (SVM), random forests (RF), gradient boosting machines (GBM), and hybrid ensemble techniques. These alternative

approaches offer distinct advantages in terms of interpretability, lower computational complexity, and suitability for smaller or structured datasets.

Moreover, it can be observed from Table 2 that only a few articles consider big data analytics. However, recent research has shown great potential in using big data analytics along with AI/ML techniques. As energy systems continue to generate massive volumes of heterogeneous and high-velocity data, the synergy between big data processing and ML techniques warrants deeper exploration. Future research should investigate the integration of ML with scalable data processing pipelines, including distributed learning, federated ML, and real-time analytics architectures. Such developments are essential for enabling the application of ML to large-scale, real-world energy datasets and ensuring the practical feasibility of ML-based decision-making in operational environments.

New and improved energy datasets

The effectiveness of ML approaches is significantly dependent on the datasets. Though there are several energy datasets available, most of them have fluctuations and irregularities [115]. These irregularities deteriorate the performance of forecasting and classification techniques. Currently, the emphasis is on improving the data processing techniques to mitigate the effects of irregularities. This leads to an exponential increase in the computational complexity of ML algorithms. Computational complexity can challenge the hardware requirements and become a bottleneck for ML-inspired implementations.

Another observation that comes directly from the standard and relevant literature is that the current datasets are more or less collected for addressing a particular set of problems. Therefore, more datasets are required that can potentially be applicable to a wide range of problems associated with energy systems. Researchers need to create more crowdsourced datasets that can be easily shared. The reason is, more often than not, energy systems require large datasets to describe the relationship between the input variables e.g. predict energy consumption, describe energy consumer behavior, etc. Crowdsourced datasets can help in gathering a large volume of unbiased data and a wide range of data in a relatively short period of time. It will not only improve the quality of the dataset but also enhances the performance of the ML-implemented solutions. To address current gaps, future research should focus on developing benchmark datasets that are multi-modal, well-annotated, and time-synchronized. In addition, there is a strong need for open-access repositories with standardized formats and documentation to enable reproducibility and cross-study comparison. Collaboration between academia, industry, and government institutions should be encouraged to generate large-scale, diverse datasets that reflect real-world operational variability across regions and energy infrastructures. Moreover, efforts should be made to incorporate uncertainty labels, noise characterizations, and missing data indicators into datasets to support the development of robust ML models that can operate under imperfect conditions.

Predictive maintenance of energy systems

Effective maintenance of complex and large-scale energy systems incurs high costs and requires sophisticated mechanisms. Predictive maintenance is the answer to this problem. With the advancements in sensing technology and ML, it has been increasingly advocated to implement predictive maintenance mechanisms for energy systems.

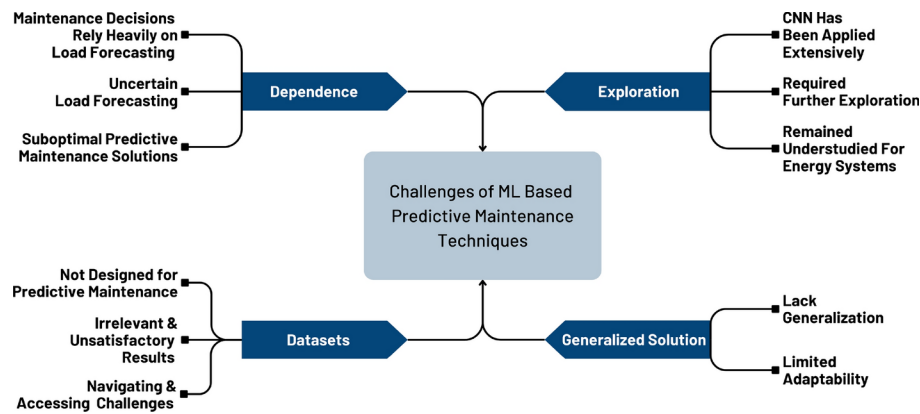


Fig. 9 Current trends and challenges for ml-based predictive maintenance techniques

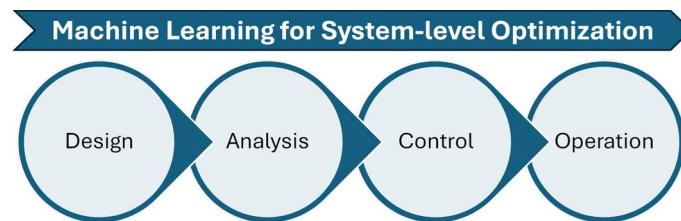


Fig. 10 System-level optimization opportunities for machine learning

Though non-ML-based techniques have been widely researched for predictive maintenance, the reliability and effectiveness of these techniques remain an open challenge. Recently, a tendency has been seen toward hybrid intelligent condition monitoring mechanisms involving both ML and non-ML-based features [116]. These mechanisms have been reported to be more accurate and robust. The current challenges facing ML-based schemes for predictive maintenance of energy systems are summarized in Fig. 9.

Optimization

Utilizing ML for optimizing energy systems is an important area of research. Optimization can be performed at various stages as presented in Fig. 10.

Optimization of the electric grid represents an important research task. It is reported in the literature that it is vital to keep recording the ratio of power utilized by the consumers and grid input power [117]. This helps in identifying the constraints of the system and thus an optimization problem covering electric grids can be formed subject to these constraints. However, due to the variability of these constraints (e.g. variability of renewable energy sources, weather, and environmental states) optimization problems are often very complex in nature and may not result in effective outputs. Due to the complex nature of these optimization problems, they are not performed regularly which leads to an even greater challenge [117]. ML approaches are not yet very popular in this area, and their application can bring much-needed adaptation to variable factors and produce more accurate results.

The last decade has produced significant literature focused on the requirement for efficient ways to produce and manage existing energy resources [92]. Therefore, optimization techniques primarily focused on bringing energy efficiency to the systems became

popular. This led to the utilization of ML approaches for energy systems. A few areas that can exploit ML for optimization are presented in Fig. 11.

Sustainability

While it is important to explore the benefits of applying ML to optimizing energy systems, it is also important to utilize the same technology for achieving sustainability goals of energy systems. Some of the examples showcasing how it could be done are briefly explained below.

ML is believed to play a vital role in bringing sustainability to energy systems. Planning is an important part of the energy system's eco cycle. Carving out additional information and meaningful insights from the energy infrastructure can aid the planning process. It is natural to use ML for this purpose. Better planning can lead to better estimates that bring sustainability to the systems [117, 118].

Energy planning for a given geographical area is critical and it is understandable that buildings and other relevant infrastructure are the biggest factors impacting energy planning. One of the sustainability goals is to have energy-efficient buildings with minimal carbon emissions. Most of the building's energy is consumed by air-conditioning (heating, cooling). A quick literature review would reveal that a significant number of works have been proposed for optimizing building energy usage. However, renewable technologies have not been much exploited for air-conditioning purposes. Therefore, there is room for exploring renewable energies specifically for air conditioning. Moreover, ML can be used to predict the energy consumption of a building for planning air-conditioning which could potentially be useful for energy-saving. It should be noted that if information on energy consumption can be obtained for most buildings in a city, then electricity transmission and distribution can be optimized as well. Future research should focus on integrating ML with passive cooling and heating systems, and on combining real-time IoT sensor data with reinforcement learning to dynamically manage HVAC loads in zero-energy buildings. In addition, simulation-based frameworks that

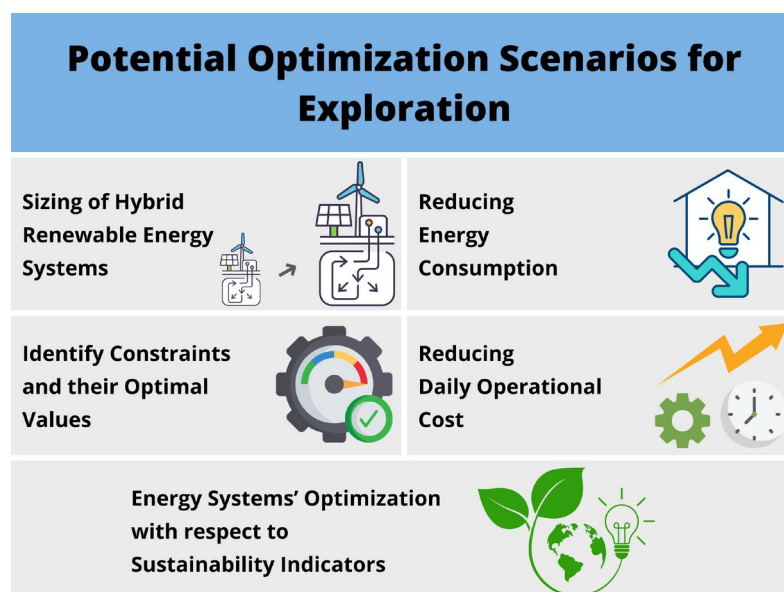


Fig. 11 ML for optimizing energy systems: avenues and opportunities

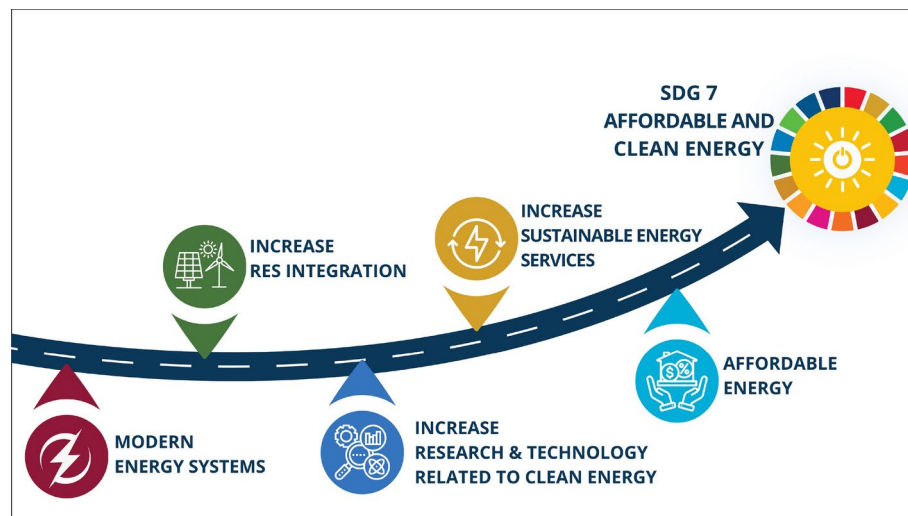


Fig. 12 The aims of the 7th sdg for energy system

use ML for city-scale retrofitting analysis should be developed to support urban energy sustainability policies.

While obtaining additional information on energy infrastructure and energy planning is important, it is imperative to bring sustainability to the materials that help build energy systems. ML can aid material engineers to synthesize materials that can bring energy efficiency to the design of photovoltaics, batteries, etc. Different ML algorithms could be used to predict and classify the properties of different materials to guide experiments and help in getting the desired output. This will bring sustainability to experimentation since the number of experiments based on the hit-and-trial approach can be significantly reduced with improved results. The 7th goal of sustainable development is dedicated to energy systems [119], its aims are summarized in Fig. 12.

Energy policy and ethics

The energy market is at a critical juncture. In the future, advancements in digital technology may have a significant impact on how we produce, distribute, and use energy. The new digitalization model relies on ML to function. Smart software that optimizes decision-making and operations will govern the autonomous integration of energy supply, demand, and renewable sources into the power grid.

Identifying research goals and designing methods to achieve these goals is significant but when it comes to implementation, a uniform policy conforming to the requirements of the technology is required. It is clear that all parts of the energy sector are advocating the shift to sustainable energy systems, however, this needs to be reinforced by consistent and strong energy policies. Governments and policy-making organizations should consider all stakeholders before announcing sustainability policies. This is important for widespread acceptability. Interestingly, even policy-making can be aided by ML since decisions are based on assumptions, trade-offs, and market trends and all these factors can be analyzed with ML which can potentially bring stability to the decision-making process [117]. The availability of ground truth i.e. supporting data is critical for developing policies. In the energy sector, there might be instances where the required data is missing. This is another potential use case of ML where missing data can be predicted.

An interesting recent study is conducted in this regard [120]. The authors elaborate on the applications of ML in predicting data useful for policy making. This study described ML as an accelerator for forming energy policies. This study exploited ML for providing useful insights relevant to energy and climate that helps in accelerating policy model development.

The utilities, power system operators, and independent power producers in the energy sector may need to put more resources into AI technology if they want to achieve substantial outcomes and maintain a competitive edge. When dealing with the difficulties of consumer safety, privacy, and adopting new business strategies, it is important to be well-informed and adaptive in regulatory engagement. Regulatory approvals for new services and goods in the new era of digital energy markets may be enforced as swiftly and efficiently as possible thanks to advancements in information technology, AI, and data analysis. In parallel, ethical challenges such as bias in ML models, lack of transparency in decision-making, and uneven access to digital infrastructure must be addressed through clear governance frameworks. Future policy research should focus on incorporating fairness metrics, explainability standards, and audit mechanisms into the regulatory structure guiding ML-based energy systems. It is also essential to develop guidelines for responsible AI deployment that align with the principles of energy justice, particularly to avoid reinforcing social inequities in vulnerable or marginalized communities.

System-level challenges and strategic research priorities

As ML integration into energy systems matures, several systemic barriers continue to limit its full-scale deployment and operational impact. These challenges are not confined to a specific energy domain but span multiple application areas and technological layers, necessitating coordinated and forward-looking research agendas.

One major challenge is the computational complexity of modern ML models, particularly deep architectures and ensemble frameworks. These models often demand significant memory, power, and processing resources, which constrain their deployment in latency-sensitive or resource-limited environments. Future research should focus on the development of lightweight ML architectures, model pruning strategies, and quantization techniques that reduce computational burden without compromising accuracy. Additionally, research into energy-efficient training algorithms and hardware-aware model design is needed to align with the sustainability goals of energy systems themselves.

Scalability and edge deployment represent another pressing concern. Most ML pipelines are developed and validated in centralized or cloud-based environments, whereas real-world energy infrastructures often require on-device inference or deployment on distributed sensor networks. There is a clear need for distributed learning frameworks, federated ML methods, and scalable model deployment protocols that support real-time operation in geographically dispersed energy systems.

Moreover, integration with legacy infrastructure remains a technical bottleneck. Existing energy control systems such as SCADA platforms are not designed to seamlessly accommodate dynamic, data-driven models. Research should prioritize the design of modular ML solutions with interoperable APIs and middleware layers that can interface with traditional systems. This includes the development of cross-platform standards,

data harmonization protocols, and plug-and-play ML modules to support easier adoption and long-term maintainability.

A related concern is interoperability, especially in multi-vendor or cross-jurisdictional energy networks. Heterogeneous hardware, data formats, and communication protocols create fragmentation that hinders cohesive ML implementation. To address this, future initiatives should establish standardized data schemas, open benchmarking environments, and conformance testing protocols for ML-driven energy applications.

Finally, explainability and trust in ML decisions remain unresolved in safety-critical energy contexts. As black-box models become increasingly prevalent, developing explainable AI (XAI) methods tailored to the energy domain is essential. This includes visualization tools, post-hoc analysis methods, and inherently interpretable model structures that can be integrated into operator workflows. Explainability is not only vital for regulatory compliance but also for building human trust and facilitating operator intervention in high-stakes scenarios.

Conclusion

Energy systems are facing new challenges such as large-scale integration of RES, demand-side management, grid management, adaption to new applications like Electric Vehicles, bringing energy efficiency to current and new systems, and achieving related SDGs. The literature suggests a growing trend to employ AI and ML approaches to address these challenges. These approaches provide befitting solutions to the complex problems of energy systems where the traditional and conventional solutions fail to provide the expected results. A comprehensive discussion of the applications of ML in the energy sector is presented. The importance, use cases, and challenges of using ML for RES are discussed as well. This work also provided a brief discussion on energy policy and energy economics. Finally, we provide detailed recommendations for consideration in future research, a brief summary is provided below.

ML-driven approaches are imperative to the advances in energy systems, however, several considerations and limitations should be respected as well. It has been observed that while other techniques remain understudied, major emphasis has been on deep learning methods. However, different applications require different AI/ML techniques and should be investigated thoroughly. Currently, research trends suggest that big data analytics is still not widely considered for energy systems. For providing better optimization strategies, data analytics should be jointly considered with AI/ML techniques. Producing sustainable energy systems conforming to the requirements of SDGs is an important goal and ML/AI techniques are believed to provide such solutions, however, significant work is required in this domain.

In addition to research directions, this work highlights several real-world implementations across the energy sector, from wind yield optimization and solar forecasting to grid behavior modeling and smart energy trading. These cases demonstrate ML's effectiveness in improving system performance, cost savings, and operational resilience. However, real-world deployment also presents critical challenges. Computational complexity and scalability limit the real-time responsiveness of ML systems, especially for large-scale operations. Forecasting uncertainty, particularly for intermittent renewables, remains a bottleneck due to underdeveloped probabilistic models. Furthermore,

integrating ML into legacy systems poses compatibility issues, and predictive maintenance systems lack standardization and robust validation benchmarks.

Ethical and governance concerns are equally important. Bias in ML training data can reinforce social and structural inequities, especially in energy justice contexts. Issues such as data privacy in smart metering, transparency of model decisions, and unequal access to enabling infrastructure must be addressed to ensure responsible and inclusive deployment of ML technologies.

Different opportunities and avenues for optimizing energy systems are highlighted in this work. Some of the avenues are still not explored and require researchers' attention. The current trends for ML-based predictive maintenance techniques are highlighted as well. Most importantly, to bring these prospects to reality, ML needs to be served with better datasets free from fluctuations and irregularities. Currently, dataset creation drives are meant to solve only a particular set of problems and thus a large volume of energy datasets that can potentially be used for various problems are required. Lastly, it should be mentioned here that all these potential research directions and recommendations are meant to exploit ML/AI approaches for green and energy-efficient systems.

Author contribution

S.A. and P.P.A. wrote the main manuscript text and prepared figures A.S.R. and A.A.M. reviewed and revised the manuscript. All authors reviewed all the versions of the manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics Approval

Not applicable.

Consent to Participate

Not applicable.

Consent to Publish

Not applicable.

Materials Availability

Not applicable.

Code Availability

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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